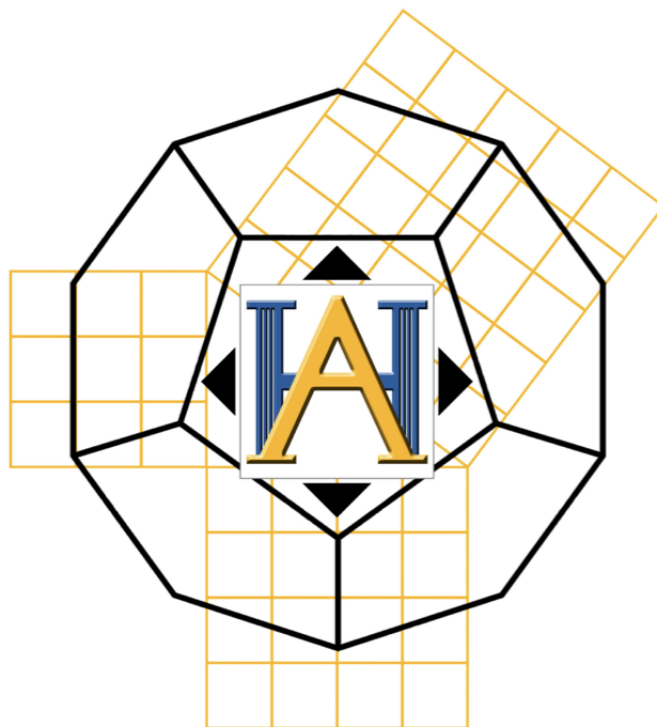




**AMERICAN HERITAGE SCHOOLS
COMPETITIVE MATHEMATICS**

2024 Mock Pre-Calculus Individual

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Instructions: The argument of a number z , $\arg z$, is equivalent to the angle θ where $z = re^{i\theta}$ for $r, \theta \in \mathbb{R}$. For a triangle, $\triangle ABC$, uppercase letters denote the angle on that vertex while lowercase letters denote the length of the side opposite to the vertex corresponding to the same letter. Denote $[ABC]$ as the area of $\triangle ABC$. NOTA means “None of the Above”. *Good luck, and have fun!*

- Compute the sum of positive integers n such that $n \leq 99$ and $\frac{99}{n} \notin \mathbb{Z}$.
 (A) 156 (B) 4695 (C) 4794 (D) 4851 (E) NOTA
- Compute $\sum_{b=4}^{16} 32_b$ in base 10.
 (A) 402 (B) 416 (C) 418 (D) 440 (E) NOTA
- Find the sum of the coefficients of $(4w - 3x + y - 3z)^{12}$ that have an x term.
 (A) -4095 (B) -1 (C) 1 (D) 4095 (E) NOTA
- How many integer ordered triples (a, b, c) are there such that $a + b + c < 22$, $0 \leq a, b, c < 14$, and $a + b + c \equiv 3 \pmod{5}$?
 (A) 240 (B) 305 (C) 350 (D) 360 (E) NOTA
- Consider the Fibonacci sequence where for all $n \in \mathbb{Z}$, $F_n = F_{n-1} + F_{n-2}$ where $F_0 = 0$ and $F_1 = 1$. Find $F_{2024} + F_{-2024} \pmod{6}$.
 (A) 1 (B) 2 (C) 3 (D) 4 (E) NOTA
- Find $\lim_{x \rightarrow 0} \frac{(1 - \cos 3x) \cdot \sin 5x \cdot \tan 4x}{x^4}$.
 (A) 0 (B) 60 (C) 90 (D) 180 (E) NOTA
- Find the sum of the slopes of the asymptotes and the abscissa and ordinate of the center of the hyperbola $(x - 4y + 2)(2x + 3y - 7) = 1$.
 (A) $\frac{7}{12}$ (B) $\frac{11}{6}$ (C) $\frac{5}{2}$ (D) $\frac{31}{12}$ (E) NOTA
- For $x \in [0, 2\pi)$, find the number of solutions to $\cos 4047x + \cos x - 2 \cos 2023x = 0$.
 (A) 2024 (B) 4047 (C) 6070 (D) 8094 (E) NOTA
- Let r_1 and r_2 be distinct roots of $x^{16} + x^{15} + \cdots + x^2 + x = 0$. What is the probability that $|r_2 - r_1| = 1$?
 (A) 0 (B) $\frac{1}{16}$ (C) $\frac{2}{17}$ (D) $\frac{1}{8}$ (E) NOTA

Use the following information for the next question:

It is actually possible to apply matrices into a variety of different functions. Take e^M , for instance, where M is a matrix. One can compute this by applying the Taylor Series for e onto M . It is also known that

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \frac{1}{1} + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

10. Using the matrix $X = \begin{bmatrix} 2 & 0 \\ 2 & 1 \end{bmatrix}$, compute e^X .

- (A) $\begin{bmatrix} e^2 & 0 \\ 2e^2 - 2e & e \end{bmatrix}$ (B) $\begin{bmatrix} e^2 & 0 \\ 2e^2 + 2e & e \end{bmatrix}$ (C) $\begin{bmatrix} e^2 & 0 \\ e^2 & e \end{bmatrix}$ (D) $\begin{bmatrix} e^2 & 0 \\ 2e^2 & e \end{bmatrix}$ (E) NOTA

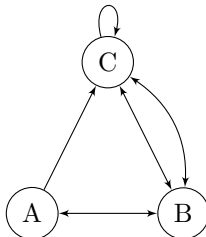
11. In $\triangle ABC$, $b = 4$ and $[ABC] = 48$. Find $a \sin C + c \sin A$.

- (A) 16 (B) 24 (C) 32 (D) 48 (E) NOTA

12. Given that $\begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{4} \end{bmatrix}$, evaluate $5 \cdot \sum_{n=0}^{\infty} \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}^n$.

- (A) $\begin{bmatrix} 6 & 4 \\ 2 & 8 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$ (C) $\begin{bmatrix} 4 & 4 \\ 3 & 6 \end{bmatrix}$ (D) $\begin{bmatrix} 5 & 4 \\ 2 & 6 \end{bmatrix}$ (E) NOTA

Use the following directed graph for the next question:



13. Consider a particle that randomly moves around this directed graph and starts at a random node. Denote a “move” to be when the particle takes any path starting at a node and ending at a node. Given that after 4 moves, the particle is at node A , how many ways can the particle end up at node B after 9 moves? Let M be the adjacency matrix of this graph. It is known that

$$M^4 = \begin{bmatrix} 7 & 18 & 24 \\ 6 & 31 & 30 \\ 12 & 24 & 37 \end{bmatrix}$$

- (A) 775 (B) 1375 (C) 2695 (D) 3637 (E) NOTA

14. Let z , z^2 , and x be colinear points in the complex plane such that $0 < \arg z < \frac{\pi}{2}$. If $|z| = 6$ and $x = 8$, find $|z^2 - z||z - x|$.

- (A) 60 (B) 112 (C) 252 (D) 282 (E) NOTA

15. Consider the Fibonacci Sequence, where $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$ and $F_0 = 0, F_1 = 1$. Compute

$$\sum_{n=3}^{\infty} \frac{2F_{n+1}}{F_n F_{n+2}}$$

- (A) $\frac{11}{12}$ (B) $\frac{16}{15}$ (C) $\frac{5}{3}$ (D) 2 (E) NOTA

16. For positive real numbers a, b such that $a > b^2$ and $ab^2 + 2b^4 = 36$, find the minimum value of

$$\frac{a^2 + ab^2 + 7b^4}{a - b^2}$$

- (A) 6 (B) $6\sqrt{3}$ (C) $8\sqrt{3}$ (D) $12\sqrt{3}$ (E) NOTA

17. Find the sum of all positive integers n such that

$$\sum_{i=1}^{1000} \lceil \log_n(i) \rceil < 997 + \sum_{i=1}^{1000} \lfloor \log_n(i) \rfloor$$

- (A) 54 (B) 55 (C) 495 (D) 496 (E) NOTA

18. Gui, Aaron, and Tiger plan on visiting the Net Café this afternoon to play *Super Smash Bros. Ultimate*. Gui and Aaron are responsible, so they arrive at the Net Café no matter what at a time between 2:00 PM and 3:00 PM; however, Tiger will only leave his home, which is 15 minutes away from the Café, either when Gui sends him a text when he arrives or at 2:30 PM (because Tiger needs his food). Each of them will leave the Café after 30 minutes. What is the probability that the 3 of them are at the Net Café at the same time and play *Super Smash Bros. Ultimate* together?

- (A) $\frac{3}{8}$ (B) $\frac{11}{16}$ (C) $\frac{23}{32}$ (D) $\frac{3}{4}$ (E) NOTA

Use the following information for the next question

The chain rule states that for a function $h(x) = f(g(x))$,

$$h'(x) = f'(g(x)) \cdot g'(x)$$

The product rule states that for $f(x) = f_1(x)f_2(x) \cdots f_n(x)$,

$$f'(x) = \sum_{i=1}^n f'_i(x) \prod_{j \neq i} f_j(x)$$

For a polynomial, $g(x) = (x - r_1)(x - r_2) \cdots (x - r_n)$,

$$g'(x) = \sum_{i=1}^n \frac{g(x)}{x - r_i}$$

19. Consider the polynomial $P(x) = R_1(x)R_2(x)R_3(x)$ where every $R_n(x)$ are linear polynomials with leading coefficient 6 for $n \in \{1, 2, 3\}$. Additionally, consider the following matrix, M .

$$M = \begin{bmatrix} R_1(x) & 0 & 0 \\ 0 & R_2(x) & 0 \\ 0 & 0 & R_3(x) \end{bmatrix} \begin{bmatrix} R_3(x) & 0 & 0 \\ 0 & R_1(x) & 0 \\ 0 & 0 & R_2(x) \end{bmatrix}$$

Find $\frac{3(\det M)'}{4\text{Tr}(M)P(x)}$.

- (A) 6 (B) 9 (C) 12 (D) 18 (E) NOTA

20. The eccentricity of a conic is defined to be the ratio between the distance of any point on the conic to the nearest focus and the perpendicular distance between that point and the nearest directrix. Consider an ellipse with center C , foci F_1 and F_2 , and directrices d_1 and d_2 , where the distance between F_1 and d_1 is less than the distance between F_1 and d_2 . Let the intersection between the line F_1F_2 and d_1 be A . Let P be an arbitrary point on the ellipse such that $PF_1 > PF_2$, $PF_1 + PF_2 = 8$, and $PF_1 = c$. Let the closest point from P on d_1 be B . Finally, $AB = 3$ and $PA \perp PF_1$. Note that $\frac{c}{a}$ is the eccentricity of this ellipse, where a is the semi-major axis. In terms of c , find the measure of $\angle PF_1F_2$.

- (A) $\arctan\left(\frac{4}{c}\right)$ (B) $\arctan\left(\frac{5}{c}\right)$ (C) $\pi - \arctan\left(\frac{4}{c}\right)$ (D) $\pi - \arctan\left(\frac{5}{c}\right)$ (E) NOTA